

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – NUMERICAL HACKATHON

Course Code:	ITA03	Course Title:	Mobile computing
CO Assessed:	CO3 – Precision (BL3)	Assessment:	Assessment Tool 3 – HACKATHON
Weightage:	30 % of CIA	Total Marks:	100
CO3 Statement:	Implement wireless networking and Mobile IP concepts for routing in cellular and ad hoc networks.. (BTL 3)		
Student Name:	Pavithra R	Reg. No.:	192421369
Section / Batch:	Slot-D	Date:	11/08/2026

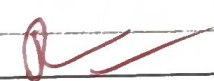
Code of Conduct

I, Pavithra.R (192421369) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: 

Instructions

- Duration: 60 minutes
- Number of questions : 10

Faculty In-charge	Course Coordinator	Head of Department
Terrance Frederick	Terrance Frederick	Terrance Frederick
Signature: <u></u>	Signature: <u></u>	Signature: _____
Date: <u>11/08/26</u>	Date: <u>11/08/26</u>	Date: _____



1. step 1: Identify the given information.

* Tunneling Over: 20 bytes per packet.

* Data packet size : 500 bytes.

step 2: Understand the goal

You need to find the additional overhead introduced by tunnel for this specific 500-byte packet.

solution:- 500 bytes of data + 20 bytes of overhead.

$$= 520 \text{ bytes}$$

1.A) The additional overhead for the 500 byte data packet is 20 bytes.

2. step 1: convert Bandwidth:

$$20 \text{ MHz} = 20 \times 10^6 \text{ Hz}$$

step 2: Find the Symbol

$$\text{symbol rate} = 20 \times 10^6 \text{ Symbol/sec}$$

step 3: calculate data rate.

Each symbol carries 6 bits.

$$\begin{aligned} \text{Data rate} &= 20 \times 10^6 \times 6 \\ &= 120 \times 10^6 \text{ bits/sec} \end{aligned}$$

$$120 \text{ Mbps}$$

QAM - Amplitude & phase

3. step 1:

no. of entries: 50

size of each entry: 32 byte

step 2: calculate the total memory
 $50 \text{ entries} \times 32 \text{ bytes/entry}$
 $= 1600 \text{ bytes}$

the total memory required is
1600 bytes or 1.6 kilo bytes.

4. step: 1

speed = 10 m/s

update interval = 2 seconds

step: 2

Formula:

Distance = speed $\times$ time

$$= 10 \times 2$$

$$= 20 \text{ meters}$$

5. step: 1

DSR, a route request packet is forwarded from one node to the next until it reaches the destination.

Source $\rightarrow$ Intermediate 1 $\rightarrow$ 2 $\rightarrow$ 3 $\rightarrow$ 4 $\rightarrow$ 5 $\rightarrow$ Destination
5 Intermediate + 1 source
 $= 6$

subnet mask: 255.255.255.240

The last octet 240 in binary is

$$240 = 11110000.$$

So the Subnet Mask is :

$$/28.$$

Step 1: Find Host bits.

IPv4 has 32 bits total.

$$32 - 28 = 4 \text{ host bits.}$$

Step 2: Find usable IP addresses

$$2^4 = 16 \text{ total addresses.}$$

But 2 Addresses are Reserved.

$\Rightarrow$ 1 network Address

$\Rightarrow$ 2 Broadcast Address

$$16 - 2 = 14.$$

$\therefore$ 14 mobile nodes can be assigned

IP address in one /28 IPv4 subnet.

$\Rightarrow$ IPv4 $\Rightarrow$ 4 parts called octets $\Rightarrow$ 8 bits.

$\Rightarrow$ subnet mask $\Rightarrow$ which part of the IP is for the Network, and which part is for devices

Step 1:

channel bandwidth = 20 MHz

Symbol Rate = 1 MSPS = 1 million Symbols/second

Step 2:

Formula:

For spectral efficiency = Data Rate / Bandwidth

Need bits per symbol to find the data rate.

Data Rate = 1 MSPS $\times$ 1 = 1 Mbps

$\therefore$ spectral efficiency = 1 Mbps / 20 MHz
= 0.05 bit/s/Hz.

$$\eta = \frac{\text{symbol rate} \times \text{Bits/symbol}}{\text{Bandwidth}}$$

If they give a modulation like 64 QAM, the bits/symbol = 6,

8.

Step 1:- No. of nodes = 50

Network area = 1 km²

transmission range = 250 m

$$\text{Network Density} = \frac{\text{No. of nodes}}{\text{Area}}$$

$$\text{Area} = \frac{\pi \times r^2}{\pi \times 250^2}$$

$$= \frac{50}{1}$$

$$= 50 \text{ nodes/km}^2$$

velocity = 300 m/s
 frequency (f) = 900 MHz (900 x 10⁶ Hz)
 speed of light (c) = A standard
 constant of 3 x 10⁸ m/s.

$$f_d = \frac{v \cdot f}{c}$$

$$f_d = \frac{300 (900 \times 10^6)}{3 \times 10^8}$$

$$f_d = \frac{27000 \times 10^6}{3 \times 10^8}$$

$$f_d = \frac{27 \times 10^9}{3 \times 10^8} = 90 \text{ Hz}$$

10.

Step 1:

Data Rate: 54 Mbps

packet size: 1500 bytes.